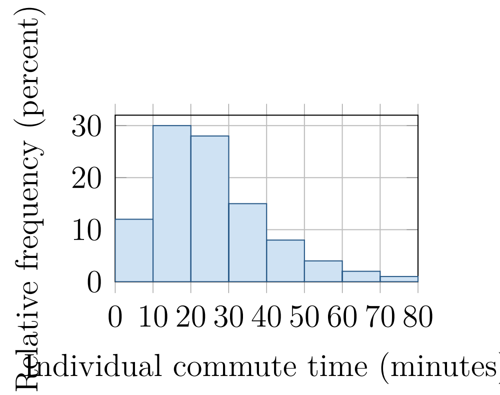
When Is the Mean Nearly Normal?

Topic 5.3 · TODAY'S PROBLEM

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: "Any average is approximately normal, even for $n = 4$."
Sol: "With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible."



FIRST TAKE (5 min, on your sheet)

For which sample size, $n = 4$ or $n = 40$, would you trust a normal model for the sample mean? Name one feature behind your choice.

DOK 1 **(a)** Identify the population shape and state the population mean and standard deviation with units.

DOK 2 **(b)** For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.

DOK 3 **(c)** Adjudicate both claims for $n = 4$ and $n = 40$. Cite the histogram shape, independence, and the CLT's sample-size requirement; use the $n = 40$ standard deviation; and explain one consequence of ignoring skew when using a normal model at $n = 4$.

Video + follow-along: `u5_lesson3_live.html` (scan the code)
Finish (a)–(c) after the video. Turn in your sheet.